

Some of the arguments in 24–32 are valid, whereas others exhibit the converse or the inverse error. Use symbols to write the logical form of each argument. If the argument is valid, identify the rule of inference that guarantees its validity. Otherwise, state whether the converse or the inverse error is made.

30. If this computer program is correct, then it produces the correct output when run with the test data my teacher gave me.

This computer program produces the correct output when run with the test data my teacher gave me.

$\therefore$ This computer program is correct.

First, we assign appropriate statement variables:

p = This computer program is correct

q = It produces the correct output

And then we can translate the arguments to logical form:

$p \rightarrow q$
 q
 $\therefore p$

Next, we can check whether the argument is valid using a truth table:

$p \rightarrow q$	q	p
T	T	T
F	F	T
T	T	F
T	F	F

By examining the critical rows, and comparing them against the conclusion, we can conclude that the argument is invalid.

And the fallacy underlying this invalid form is called the converse error because the conclusion of the argument would follow from the premises if the premise were $p \rightarrow q$ were replaced by its converse: $q \rightarrow p$.

This replacement is not allowed, because a conditional statement is not logically equivalent to its converse, as proven below:

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Solution: The argument form is invalid, and the fallacy is called the converse error.